CS328 A2 — Writeup

Most important decision. Going back to the three axes after the magnitude-only features stalled at 57%. A vertical proxy (motion projected on its main axis) and the residue Sensor Logger's gravity removal leaves on each axis raised accuracy on unseen recordings to 65.7% (recordings: 66.8%). Judging every choice with whole recordings held out, not the leaky window split, is what told us which changes were real.

Biggest surprise in the confusion matrix. Lying is called sitting 59% of the time and is right only 25% of the time. Looking at the traces, about half the lying and sitting recordings are a phone resting completely still, so they contain no posture information at all, and no model can fix that.

Next. Log the gravity vector and barometer, add tilt and altitude-change features for lying/sitting and stairs, and evaluate leave-one-person-out once participant IDs are available.

Contribution statement. Harish Balakrishnan: set up the group repo and uploaded the starter notebook, util.py and class data. Vedant Naidu: data loading and windowing, feature engineering, model comparison and tuning, failure analysis, exploration report and writeup.